Mechanics book

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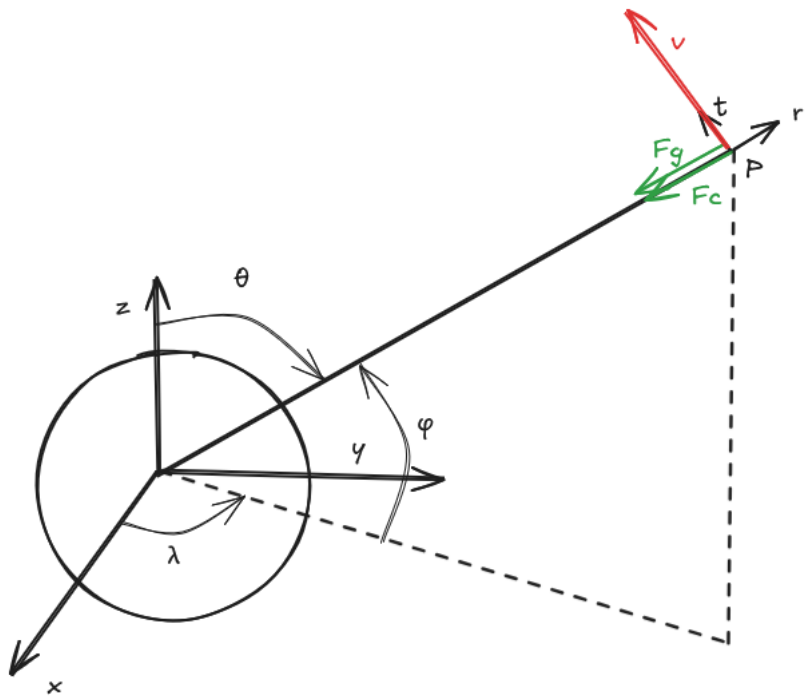
1 Satellite orbit

1.1 Introduction

Study how to maintain a satellite at a certain orbit

1.2 Schema

A satellite of a mass m is orbiting around a planet of mass M at altitude ρ [meters] from its center with a tangential speed of v m/s.



The satellite position in a spherical referential (ρ, θ, λ) is :

$$\vec{OS} = \begin{bmatrix} l \\ m \\ n \end{bmatrix}_{R1} = \rho \cdot \begin{bmatrix} \sin(\theta) \cdot \cos(\lambda) \\ \sin(\theta) \cdot \sin(\lambda) \\ \cos(\theta) \end{bmatrix}_{R1} \tag{1}$$

with

- ρ : radial distance from the center of the planet (origin) to the object, in meters.
- λ : longitude, angle of rotation on the z axis
- ϕ : latitude of the satellite, elevation angle from the plane formed by x and y axes. (elevation angle, altitude angle)
- θ : polar angle between the polar axis(here the z axis) and the radial distance to the satellite. Or angle between the zenith and the latitude of the satellite at altitude ϕ Complementary to ϕ. (inclination angle, zenith angle,) $\theta = \frac{\pi}{2} - \phi$
- λ : longitude, which is the azimuthal angle, which angle of rotation around the polar axis

These angles are chosen to match the angles used for GPS location on Earth : latitude ϕ and longitude λ
TODO : express the velocity vector of the satellite

1.3 Stable orbit

1.3.1 Problem

An artificial object (i.e a the satellite or a space craft)is orbiting around a space body at speed v and at altitude h_{sat}. Calculate the force needed to maintain the object at that specific altitude

1.3.2 Resolution

Without any force applied on the artificial object, it will continue moving in the same direction as the speed vector. To maintain it constant altitude, i.e. forcing a circular trajectory around the planet, a centripetal force is applied toward the space body (S) center :

$$F_c = \frac{m_{sat} \cdot v_t^2}{\rho} = \frac{m_{sat} \cdot v_t^2}{S_{radius} + h_{sat}} \tag{2}$$

$$F_c = m_{sat} \cdot \omega^2 \cdot \rho = m \cdot (\frac{2\pi}{T})^2 \cdot (S_{radius} + h_{sat}) \tag{3}$$

Todo : create table at the beginning describing each variables and its unit,
m_{sat} : mass of the orbiting object, in kg
ρ : distance in m from the center of the rotating object (planet, sun, moon) to the orbiting object
v_t : tangential velocity of the orbiting object, in m/s
S_{radii} : Space body radii, in m
h_{sat} : distance from the surface of the space body to the satellite, in m
$\omega = \frac{2\pi}{T}$: angular speed of the space body, in rad/s
T : orbital period, in s

In space, there are several ways to apply this force, like :

- gravity force, if the orbiting object is within the gravity field of the space body.
- thrusts applying force equal to F_c at given time

CASE A : Gravity force
A.1. compute spped to stay at a given altitude
A.2. compute geostationary altitude, for different planets (check cases where the space body is gas and the satellite must orbit inside it)

CASE B : thrusters force :
compute fuel needed to maintain this (like for the ISS)

1.3.3 Case A : Satellite within space body's gravity field

$$F_c = m_{sat} \cdot g \tag{4}$$

$$m_{sat} \cdot \frac{v_t^2}{\rho} = m_{sat} \cdot \omega^2 \cdot \rho = m_{sat} \cdot g \tag{5}$$

$$\frac{v_t^2}{\rho} = \omega^2 \cdot \rho = g = \frac{G \cdot M_S}{\rho^2} \tag{6}$$

$$v_t = \sqrt{\frac{G \cdot M_S}{\rho}} \tag{7}$$

$$\omega = \sqrt{\frac{G \cdot M_S}{\rho^3}} \tag{8}$$

$$\rho = \frac{G \cdot M_S}{v_t^2} \tag{9}$$

$$\rho = \sqrt[3]{\frac{G \cdot M_S}{\omega^2}} \tag{10}$$

choose v_t (or ω) to stay at a specific orbit. Or if the orbiting object must have a specific speed, the altitude that it must reach is computed. Then, it is possible to check if the orbit at that altitude is free and can be used without colliding with other objects.

1.3.4 Geostationary orbit

Orbit where an object is moving at the same speed as the planet, and is always in the same position on the sky. if it is satellite used for observation, it will look at exactly the same location on Earth. Some satellite follow this orbit, mainly for international communication. to maintain a the orbiting object, the object has the same angular speed as the space body $\omega = \omega_S$

$$\omega = \omega_S \tag{11}$$

$$\rho_{go} = \sqrt[3]{\frac{G \cdot M_{SB}}{\omega_S^2}} \tag{12}$$

$$v_{go} = \omega_S \cdot r_{go} \tag{13}$$

Example for Earth : Period T of 23h, 56min and 4s

$$T = 24 * 3600 + 56 * 60 + 4 \tag{14}$$

$$\omega = \omega_{Earth} = \frac{2 \cdot \pi}{T} = \frac{2 \cdot \pi}{23h56min4s} \tag{15}$$

$$\rho_{go} = \sqrt[3]{\frac{G \cdot M_{SB}}{\omega^2}} \tag{16}$$

$$v_{go} = \omega_S \cdot r_{go} \tag{17}$$

As long as the orbiting object maintain this speed, it will stay at this orbit. However, friction with the high atmosphere might deviate the satellite from its path. some of them are equipped with thrusters to correct the trajectory when needed.
Application example : - telecommunication satellites, placed around the planet to ensure that it is fully covered - observation satellite : some satellites are placed to observe the same points of interest

1.3.5 Case B : Orbit outside gravity field

Altitude when gravity is very low

$$g = \frac{G \cdot M_S}{\rho^2} <= 0.001 \tag{18}$$

$$\rho > \sqrt{\frac{G \cdot M_S}{g}} \tag{19}$$

Satellite is attracted by other space bodies, so thrusters and or boosters are needed to maintain the expected path.

$$\vec{v} = \begin{bmatrix} v_\rho \\ v_\theta \\ v_\lambda \end{bmatrix}_{R1} \tag{20}$$

The components v_θ and v_λ depends on the v need to stay on orbit will maintain the satellite at orbit.

$$v = \sqrt{v_\theta^2 + v_\lambda^2} \tag{21}$$

The component v_ρ must be compensated by activating thrusters and boosters to keep the altitude ρ at the same value.

1.3.6 les constantes

- G :
- altitude de la position de départ

- k : coefficient caractéristique de la géométrie du solide
- C_x
- S : surface à l'air
-

$$\vec{B_1B_2}_{R_1} = \begin{bmatrix} v_1 \cdot \cos(\beta_4) \\ 0 \\ v_1 \cdot \sin(\beta_4) \end{bmatrix}_{R1} \quad \vec{B_1C_1}_{R_1} = \begin{bmatrix} c_1 \cdot cos(\beta_1) \\ 0 \\ c_1 \cdot sin(\beta_1) \end{bmatrix}_{R1} \quad \vec{C_1C}_{R_1} = \begin{bmatrix} c \cdot cos(\beta_1) \\ 0 \\ c \cdot sin(\beta_1) \end{bmatrix}_{R1} \tag{22}$$

$$\vec{C_1C_2} = \begin{bmatrix} v_2 \cdot cos(\gamma_1) \\ 0 \\ v_2 \cdot sin(\gamma_1) \end{bmatrix}_{R1} \quad \vec{CC_2}_{R_1} = \begin{bmatrix} -c_2 \\ 0 \\ 0 \end{bmatrix}_{R2} \quad \vec{C_2D}_{R_2} = \begin{bmatrix} -d \\ 0 \\ 0 \end{bmatrix}_{R2} \quad \vec{DD_1}_{R_3} = \begin{bmatrix} -d_1 \\ 0 \\ 0 \end{bmatrix}_{R_3} \tag{23}$$

$$\vec{D_1D_2}_{R_3} = \begin{bmatrix} -d_2 \cdot cos(\delta_1) \\ 0 \\ d_2 \cdot sin(\delta_1) \end{bmatrix}_{R_3} \tag{24}$$

-
-
-

L : longueur représentative de la fusée

$$ggagsfaaf \tag{25}$$

1.4 Formules utiles

Reynolds number

$$Re = \frac{\rho_{air} \cdot L \cdot v_{fusee}}{\mu} \tag{26}$$

accélération terrestre

$$g = G\frac{m_{terre}}{(rayon_terre + altitude_{z_0} + z)^2} \tag{27}$$

Force de pesanteur

$$Fg = m_f usee \cdot g \tag{28}$$

frottement laminaire

$$Ffl = -k * \eta * v_{fusee} \tag{29}$$

frottement turbulant

$$Fft = -\frac{1}{2} \cdot \rho_{air} \cdot C_x \cdot S \cdot v_{fusee}^2 \tag{30}$$

quantité de mouvement conservée

$$p_{fusee} = p_{gaz} \tag{31}$$
$$P_{fusee} = m_{fusee} * v_{fusee} \tag{32}$$
$$P_{gaz} = m_{gaz} * v_{gaz} \tag{33}$$
$$dp/dt = sommeF = m_{fusee} * a_{fusse} + dm_{fusee}/dt + v_{fusee} \tag{34}$$
$$dp/dt = m_g az * a_g az + dm_g az/dt * v_{gaz} = F_{poussee} \tag{35}$$

equilibre des forces

$$\sum F = F_{poussée} - F_g - F_{frottement} \tag{36}$$
$$a = \frac{\sum F}{m_{fusee}} \tag{37}$$

Les formules standards de la cinématique donnent vitesse et position.

$$v_{fusee} = a \cdot \Delta t \tag{38}$$
$$z = a \cdot \Delta t^2 + v_{fusee} \cdot \Delta t + z_0 \tag{39}$$

1.5 Formules utiles

nombre Reynolds

$$Re = \frac{\rho_{air} \cdot L \cdot v_{fusee}}{\mu} \tag{40}$$

accélération terrestre

$$g = G \frac{m_{terre}}{(rayon_t erre + altitude_{z_0} + z)^2} \tag{41}$$

Force de pesanteur

$$Fg = m_f usee \cdot g \tag{42}$$

frottement laminaire

$$Ffl = -k * \eta * v_{fusee} \tag{43}$$

frottement turbulant

$$Fft = -\frac{1}{2} \cdot \rho_{air} \cdot C_x \cdot S \cdot v_{fusee}^2 \tag{44}$$

quantité de mouvement conservée

$$p_{fusee} = p_{gaz} \tag{45}$$
$$P_{fusee} = m_{fusee} * v_{fusee} \tag{46}$$
$$P_{gaz} = m_{gaz} * v_{gaz} \tag{47}$$
$$dp/dt = sommeF = m_{fusee} * a_{fusse} + dm_{fusee}/dt + v_{fusee} \tag{48}$$
$$dp/dt = m_gaz * a_gaz + dm_gaz/dt * v_{gaz} = F_{poussee} \tag{49}$$

equilibre des forces

$$\sum F = F_{poussée} - F_g - F_{frottement} \tag{50}$$
$$a = \frac{\sum F}{m_{fusee}} \tag{51}$$

Les formules standards de la cinématique donnent vitesse et position.

$$v_{fusee} = a \cdot \Delta t \tag{52}$$
$$z = a \cdot \Delta t^2 + v_{fusee} \cdot \Delta t + z_0 \tag{53}$$

1.6 Procédure de calcul

itérer à chaque pas de temps
calculer Re selon valeur Re, le frottement est laminaire ou turbulant. limite environ vers 3000
force gravité les frottements équilibre des forces accélération pas de temps vitesse positition

1.7 Calcul plus poussé

masse variable quantité de mouvement et poussée des gaz. perte de masse à cause des gaz
g change selon l'altitude
les propriétés de l'air change, ρ, μ, η, ...
L de la fusée si elle perd des étages
calcul précis du C_x : cf TAero, ou cours mécanique fluide incompressible EPFL

1.8 Conclusion

1.9 Other cases

1. establish the equation of the general movement of a satellite (the same as on Earth)
Acceleration

$$\vec{a} = \frac{\sum \vec{F}}{m_{sat}} \tag{54}$$

Force equilibrium in a spherical referential (check if this is correct. maybe another force must be added. Centripetal force is only tangential)

$$\sum \vec{F} = \vec{F_c} - \vec{F_g} = \vec{F_R} = \begin{bmatrix} F_c \cdot cos(\alpha) \\ F_c \cdot sin(\alpha) \\ 0 \end{bmatrix}_{R1} + \begin{bmatrix} -F_g \\ 0 \\ 0 \end{bmatrix}_{R1} = m_{sat} \cdot \begin{bmatrix} a_r \\ a_t \\ a_z \end{bmatrix}_{R1} \tag{55}$$

$$F_c \cdot cos(\alpha) = m_{sat} \cdot g \tag{56}$$

$$m_{sat} \cdot \frac{v_t^2}{r} \cdot cos(\alpha) = m_{sat} \cdot \omega^2 * r \cdot cos(\alpha) = m_{sat} \cdot g \tag{57}$$

$$\frac{v_t^2}{r} \cdot cos(\alpha) = \omega^2 \cdot r \cdot cos(\alpha) = g = \frac{G \cdot m_{Earth}}{r^2} \tag{58}$$

$$v_t = \sqrt{\frac{G \cdot m_{Earth}}{r \cdot cos(\alpha)}} \tag{59}$$

$$\omega = \sqrt{\frac{G \cdot m_{Earth}}{r^3 \cdot cos(\alpha)}} \tag{60}$$

$$\tag{61}$$

2. Drag and lift forces depend if there is an atmosphere or not at that altitude -> compute air density depending on altitude. it will become 0 in space 3. simulate different cases with different propulsion, ...
4. test case : propulsion force that must be applied to maintain orbit over the planet (Earth)
5. the same case but using gravity force
6. so the same calculations for all planets and moon in the solar system
7. ignore sun heat and radiation, do the same with the sun
8. other case : how close a satellite be from the sun ? can a satellite orbit around it ? will it susstain the heat, the radiation form the sun ?

2 Putting a rocket in orbit

2.1 Introduction

Etudier la cinématique du lancement et de la mise en orbite d'une fusée

la mise en orbite sera faite en prenant en compte les aspects dynamiques

2.2 Schema

schéma représentatif de la situation
schéma de l'atmosphère, avec formules de températures et pression selon altitude

2.3 Etude cinématique

2.3.1 les constantes

- G :
- altitude de la position de départ
- k : coefficient caractéristique de la géométrie du solide
- C_x
- S : surface à l'air
-

$$\vec{B_1B_2}_{R_1} = \begin{bmatrix} v_1 \cdot cos(\beta_4) \\ 0 \\ v_1 \cdot sin(\beta_4) \end{bmatrix}_{R1} \quad \vec{B_1C_1}_{R_1} = \begin{bmatrix} c_1 \cdot cos(\beta_1) \\ 0 \\ c_1 \cdot sin(\beta_1) \end{bmatrix}_{R1} \quad \vec{C_1C}_{R_1} = \begin{bmatrix} c \cdot cos(\beta_1) \\ 0 \\ c \cdot sin(\beta_1) \end{bmatrix}_{R1} \tag{62}$$

$$\vec{C_1C_2} = \begin{bmatrix} v_2 \cdot cos(\gamma_1) \\ 0 \\ v_2 \cdot sin(\gamma_1) \end{bmatrix}_{R1} \quad \vec{CC_2}_{R_1} = \begin{bmatrix} -c_2 \\ 0 \\ 0 \end{bmatrix}_{R2} \quad \vec{C_2D}_{R_2} = \begin{bmatrix} -d \\ 0 \\ 0 \end{bmatrix}_{R2} \quad \vec{DD_1}_{R_3} = \begin{bmatrix} -d_1 \\ 0 \\ 0 \end{bmatrix}_{R_3} \tag{63}$$

$$\vec{D_1D_2}_{R_3} = \begin{bmatrix} -d_2 \cdot cos(\delta_1) \\ 0 \\ d_2 \cdot sin(\delta_1) \end{bmatrix}_{R_3} \tag{64}$$

3 Kinematics

3.1 Introduction

The objective of this report is to compute the trajectory of a projectile following the ballistic mechanics.

3.2 Formulas

$$\sum F = m \cdot a(t) \tag{65}$$

$$a(t) = \frac{\sum F}{m} \tag{66}$$

$$A(t) = \int a(t) \cdot dt \rightarrow \tag{67}$$

$$v(t) = \int_{t_1}^{t_2} a(t) \cdot dt = A(t_2) - A(t_2) \tag{68}$$

$$x(t) = \int v(t)dt = \iint a(t)dt^2 \tag{69}$$

$$\tag{70}$$

case a and v constant

$$v(t) = \int a \cdot dt = a \cdot t + C \tag{71}$$

$$x(t) = \iint a \cdot dt^2 = \int (a \cdot t + C)dt = \frac{1}{2}a \cdot t^2 + C \cdot t + D \tag{72}$$

Using the initial condition for x(t) and v(t), we find :

$$v(t) = a \cdot t + v_0 \tag{73}$$

$$v(t_2) - v(t_1) = \int_{t_1}^{t_2} a(t) \cdot dt = (a \cdot t + v_0)|_{t_1}^{t_2} \tag{74}$$

$$x(t) = \frac{1}{2}a \cdot t^2 + v_0 \cdot t + x_0 \tag{75}$$

$$x(t_2) - x(t_1) = (\frac{1}{2}a \cdot t^2 + v_0 \cdot t + x_0)\Big|_{t_1}^{t_2} \tag{76}$$

Impulse is the variation of momentum, when integrating force in respect to time :

$$I = \int F \cdot dt = \int m(t)a(t) \cdot dt = \Delta p = p_2 - p_1 \tag{77}$$

Simple case where the mass is known :

$$I = \int m \cdot a(t) \cdot dt = m \cdot v + C = \Delta p = m_2 \cdot v_2 - m_1 \cdot v_1 \tag{78}$$

Kinetic energy : integral of force with respect to distance

$$a = dv/dt = dv/dx \cdot dx/dt = dv/dx \cdot v \tag{79}$$

$$\int F \cdot dx = \int madx = \int m \cdot v \cdot \frac{dv}{dx} dx = \int m \cdot v \cdot dv = \frac{1}{2} \cdot m \cdot v^2 + C \tag{80}$$

$$\Delta KE = \frac{1}{2} \cdot m \cdot v^2 + C = \frac{1}{2} \cdot m \cdot v_2^2 - \frac{1}{2} \cdot m \cdot v_1^2 \tag{81}$$

In many complex systems, the mass is not constant, for

- Rocket propulsion
- Driving vehicle

In these cases, the mass decreases as the fuel is used. We need to consider that the mass depends on time $m = m(t)$. The object position can be calculated using the time $x(t)$. From this function, we can link the mass m to the distance $m(t)$ -> $m(x) = m(x(t))$, in order to compute the kinetic energy.

$$I = \int F \cdot dt = \int m(t)a(t) \cdot dt \tag{82}$$

$$\Delta KE = \int F \cdot dx = \int m(x) \cdot a(x) \cdot dx = \tag{83}$$

3.3 Examples

Hitting moving target with unguided rockets

3.4 Initial conditions

projectile initial positions :

$$px = px_0 + pv * cos(p_{alpha}) * \Delta t \tag{84}$$

$$py = py_0 + pv * sin(p_{alpha}) * \Delta t \tag{85}$$

Target positions :

$$Tx = Tx_0 + Tv * cos(T_{alpha}) * \Delta t \tag{86}$$

$$Ty = Ty_0 + Tv * sin(T_{alpha}) * \Delta t \tag{87}$$

The target is hit when :

$$Tx = px \tag{88}$$

$$Tx_0 + Tv * cos(T_{alpha}) * \Delta t = px_0 + pv * cos(p_{alpha}) * \Delta t \tag{89}$$

$$(Tx_0 - px_0) + [Tv * cos(T_{alpha}) - pv * cos(p_{alpha})] * \Delta t = 0 \tag{90}$$

$$Ty = py \tag{91}$$
$$Ty_0 + Tv * sin(T_{alpha}) * \Delta t = py_0 + pv * sin(p_{alpha}) * \Delta t \tag{92}$$
$$(Ty_0 - py_0) + [Tv * sin(T_{alpha}) - pv * sin(p_{alpha})] * \Delta t = 0 \tag{93}$$

$$(Tx_0 - px_0) + [Tv * cos(T_{alpha}) - pv * cos(p_{alpha})] * \Delta t = 0 \tag{94}$$
$$(Ty_0 - py_0) + [Tv * sin(T_{alpha}) - pv * sin(p_{alpha})] * \Delta t = 0 \tag{95}$$

In this problem, we have 2 unknown variables : Δt and p_{alpha}.
Launching angle : $alpha$
Propulsion force : F_p
Gravity force : F_g
If the projectile is launched from a canon, friction with cannon : $F_{friction}$ We will discuss this case at the end of the report.

3.5 Reynolds Number

The Reynolds number is an non-dimensional number, and used to define if the fluid flow is laminar or turbulent.

$$Re = \frac{\rho \cdot u \cdot L}{\mu} \tag{96}$$

ρ : density of the fluid
u : flow speed
L : characteristic linear dimension
μ : dynamic viscosity of the fluid

The characteristic linear dimension L depends on the shape of the object of study. Here some example :

- Plane wing : length of the wing
- Hydraulic pipe : diameter of the pipe
- Complex shape : the biggest dimension

Depending of the result, the flow can have 3 regimes

- If $Re <$ 2040, the flow is still considered laminar

- If Re > 2100, the flow is turbulent
- If 1800 < Re < 2100, the flow is in the transition/intermediary range, which is a mix of laminar and turbulent [1]

For each regime, the drag force is different. For turbulent and laminar regimes, the formula is as follows:

$$F_d^{turbulent} = \frac{1}{2} \cdot \rho \cdot C_D \cdot S \cdot u^2 \tag{97}$$

$$F_d^{laminar} = C_F \cdot \rho \cdot D^2 \cdot u^2 \tag{98}$$

S: cross frontal section
C_D : turbulent drag coefficient
C_F : laminar drag coefficient

The intermediary regime is a mix from the 2 base regimes. In this situation, an approximation can be evaluated by computed the percent Re compared to 1800 and 2100.

$$F_d^{transition} = (1 - \frac{Re - 1800}{2100 - 1800}) \cdot F_d^{laminar} + \frac{Re - 1800}{2100 - 1800} \cdot F_d^{turbulent} \tag{99}$$

3.6 Cannon cinematic

In this case, we need to consider what happens to the projectile inside the barrel, from the moment the projectile starts moving until it exits the barrel.
The projectile is launched by the mean of an explosive charge placed in the breach. When the shooter presses on the fire button, this charge is set off.
The energy of the charge is used as propulsion energy for the projectile. A small portion of the energy is used to spin the projectile, which will stabilize the trajectory during the flight.

$$E_{charge} = E_{kin}^{translation} + E_{kin}^{rotation} \tag{100}$$

$$E_{charge} = \frac{1}{2} \cdot m_{projectile} \cdot v^2 + \frac{1}{2} \cdot I \cdot \omega^2 \tag{101}$$

To push the projectile, the energy of the charge will generate a pressure. Using the definition of the pressure and energy as functions of force, it is possible to define the pressure as function of energy.

[1] https://en.wikipedia.org/wiki/Laminar_flow

$$p = F/A \tag{102}$$

$$E = F \cdot d \tag{103}$$

$$p = \frac{F}{A \cdot d} = \frac{E}{V} \tag{104}$$

In our example, the energy used to move the projectile is the kinetic energy of translation $E_{kin}^{translation}$, F is the propulsion force F_p, and V is the initial volume of the exploded charge (gas).

$$p_{charge} = \frac{F_p}{A \cdot d} = \frac{E_{kin}^{translation}}{V_{charge}} \tag{105}$$

3.6.1 Charge/explosion volume

$$V_{charge}^{inital} = A_{projectile} \cdot h_{projectile} = \frac{D_{projectile}^2}{4} \cdot h_{projectile} \tag{106}$$

After the charge is set off, this volume will change

$$V(t) = V_{charge}^{inital} + Position_{projectile}(t) \cdot A_{projectile} \tag{107}$$

3.6.2 Propulsion pressure

In reality, we need to consider the atmospheric pressure.

$$P_{charge} - p_{atm} = \frac{F_p}{A \cdot d} = \frac{E_{kin}^{translation}}{V_{charge}} \tag{108}$$

After the charge is set off, the projectile starts moving by the action of P_{charge} on its section $A_{projectile}$, which induces the propulsion force F_p.

$$F_p = P_{charge} \cdot A_{projectile} = P_{charge} \cdot \pi \cdot \frac{D_{projectile}^2}{4} \tag{109}$$

As the projectile is moving away from the breech, V_{charge} will increase and thus the pressure is reduced. Consequently, the pressure will keep pushing the projectile, but with lower propulsion force. This ends with one of these situations happens:

- The projectile is out the barrel
- P_{charge} is equal to p_{atm}

3.6.3 Projectile spinning

To stabilize its trajectory during its flight and increase the range, the projectile should spin with a specified angular speed. This is achieved by the thread inside the barrel. The mathematics formula is as follows :

$$E_{kin}^{rotation} = \frac{1}{2} \cdot I \cdot \omega^2 \tag{110}$$

The spinning will produce an angular momentum L:

$$L = m \cdot v \cdot r \tag{111}$$

where, m : is the projectile mass v : projectile velocity r : projectile radium
The rotational kinetic energy is the highest the moment the projectile exits the barrel. During the flight and due to the friction with air, the spinning is reduced. this value should be optimized because - to produce this spinning, the projectile has friction with the internal diameter of the barrel. The threads will initiate the projectile rotation.
the more the friction, the higher is the spinning, but it will also reduce the kinetic energy, -> lower velocity
if the friction is not enough, rotational energy is low, the trajectory is not stabilized enough and the accuracy of the shot is low.
For this, test with different - spinning thread shape and number - barrel length - explosive charge.
The angular momentum must be optimized.
To make sure it is optimized, the shell is placed on a test bench, submitted to a wind flow. This simulated the shell during the flight. in the same time, the projectile is spinning. the wind flow is reduced to simulate the air that slows the energy of the projectile. during the simulation, we measure the angular momentum and its resulting force to find if it will stays on the computed trajectory. the angular speed is controlled at the beginning of the simulation, when the shell exits the barrel.

table

Table 1: Holes and threaded holes

Screw	Threaded hole diameter	Bore Hole (clearance)
M3	2.8	3.2
4	35	144
5	45	300

Clearance To fix two part together, a clearance is needed. When two parts must be assembled together, a clearance of 0.1mm is enough. Then use the glue to fix the assembly.

Table 2: Nonlinear Model Results

Case	Method#1	Method#2	Method#3
1	50	837	970
2	47	877	230
3	31	25	415
4	35	144	2356
5	45	300	556

$$Length_{lattitude}(\phi) = 111132.92 - 559.82 \cdot cos(2 \cdot \phi) + 1.175 * cos(4 \cdot \phi) - 0.0023 \cdot cos(6 \cdot \phi) = ...[m/degree] \tag{112}$$

1 degree longitude at latitude phi

$$Length_{longitude}(\phi) = 111412.84 - 93.5 \cdot cos(3 \cdot \phi) + 0.118 \cdot cos(5 \cdot \phi) = ...[m/degree] \tag{113}$$

3.7 Collisions

https://www.youtube.com/watch?v=M2xnGcaaAi4

3.7.1 Elastic collision

Momentum is conserved

$$\sum_i \vec{p_i} = \vec{P} \tag{114}$$

$$m_1 \cdot \vec{v_{1i}} + m_2 \cdot \vec{v_{2i}} = m_1 \cdot \vec{v_{1f}} + m_2 \cdot \vec{v_{2f}} \tag{115}$$

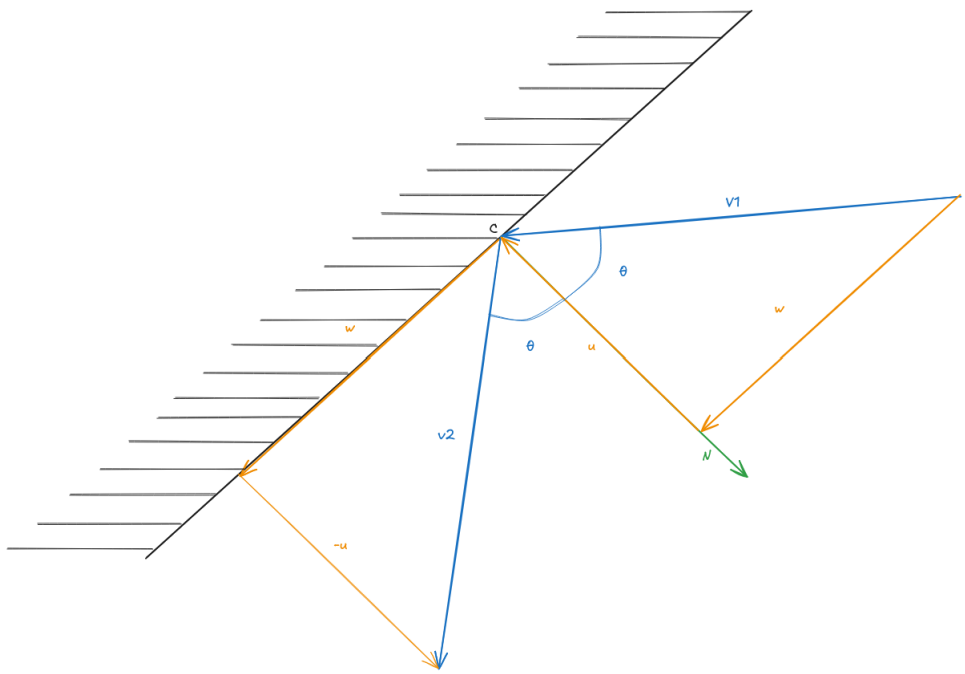
Energy is conserved

$$E^i_{kin} = E^f_{kin} = \frac{1}{2} \cdot m \cdot v_1^2 = \frac{1}{2} \cdot m \cdot v_2^2 \tag{116}$$

3.7.2 Pool balls

3.7.3 Ricochet

An object is bouncing off a solid surface.
The object collides on the surface with an angle θ to the normal vector and will bounce off with the same angle.



We consider a perfect elastic collision. - bodies in collision stay separate after - No Energy losses : Energy is conserved

$$E_{kin}^{i} = E_{kin}^{f} = \frac{1}{2} \cdot m \cdot v_1^2 = \frac{1}{2} \cdot m \cdot v_2^2 \tag{117}$$

Velocity vector can be projected on the surface and its normal :

$$\vec{v_1} = \vec{u} + \vec{w} \tag{118}$$

$\vec{u}$ is the projection of $\vec{v_1}$ on the surface normal $\vec{n}$

$$\vec{u} = \frac{\vec{v_1} x \vec{n}}{\vec{n} x \vec{n}} \vec{n} \tag{119}$$

$$\vec{w} = \vec{v_1} - \vec{u} \tag{120}$$

where x is the scalar product.

$$\vec{v_2} = \vec{w} - \vec{u} \tag{121}$$

3.7.4 Near elastic collision

bodies stay separate, energy is lost to sound and heat

$$E_{kin}^{i} = E_{kin}^{f} + E_{heat} + E_{sound} \tag{122}$$

3.7.5 Plastic collision

3.7.6 Perfect Inelastic collision

Energy is also lost in deformation, like 2 cars in collision and stay together after Momentum

$$m_1 \cdot v_1 + m_2 \cdot v_2 = (m_1 + m_2) \cdot v_f \tag{123}$$

$$E_{kin}^{i} = E_{kin}^{f} + E_{heat} + E_{sound} + E_{deformation} \tag{124}$$

3.8 Schéma cinématique

3.9 Vecteurs positions

origine : centre de rotation verticale se trouvant sous les pâles principales.

position des pâles principales (pp) : vecteur verticale

position de l'hélice arrière : vecteur allant de l'origine vers l'hélice (h) arrière.

3.10 Angular momentum

3.10.1 Formula

$$\vec{L} = \vec{OA} \otimes \vec{P} = \vec{r} \otimes \vec{P} = \vec{r} \otimes m \cdot \vec{v} = \vec{I} \otimes \vec{\omega} \tag{125}$$

$\vec{L}$: Angular Momentum [$kg \cdot \frac{m^2}{s}$]
$\vec{OA}$ and r: position of the mass [m] according to a referance
$\vec{P}$: linear momentum [$kg \cdot \frac{m}{s}$][2]
$\vec{v}$: velocity [$\frac{m}{s}$] I : moment of inertia [$m^2 \cdot kg\cdot$]
ω : angular speed [$\frac{rad}{s}$]
Torque :

$$M = \frac{d\vec{L}}{dt} = \frac{d(\vec{I} \otimes \vec{\omega})}{dt} \tag{126}$$

if we consider a particule of mass m, $\vec{r}$ is the position of the center of mass. If it is a solid object, L is first computed according to the axis of rotation of the object :

$$\vec{L}_{ar} = \vec{I}_{ar} \otimes \vec{\omega}_{ar} \tag{127}$$

To compute the angular moment according to an other axis of rotation (new referance), we use the Huygens-Steiner theorem (or the Parallel axis theorem) :

$$\vec{L}_0 = \vec{I}_0 \otimes \vec{\omega}_{cm} \tag{128}$$

$$\vec{I}_0 = \vec{I}_{ar} + m \cdot d^2 \tag{129}$$

with d the distance between the axis of rotation of the object and the new referance.

3.10.2 Condition of stability

Main rotor(s):

$$\vec{L}_{mr} = \vec{r}_{mr} \otimes m_{mr} \cdot \vec{v_{mr}} = \vec{I_{mr}} \otimes \vec{\omega_{mr}} \tag{130}$$

Rear rotor :

$$\vec{L}_{rr} = \vec{r}_{rr} \otimes m_{rr} \cdot \vec{v_{rr}} = \vec{I_r r} \otimes \vec{\omega_{rr}} \tag{131}$$

assurer la stabilité lors du vol: les moments cinétiques doivent s'annuler. (poser la formule et résoudre)

$$\vec{L_{mr}} = \vec{L_{rr}} \tag{132}$$

or
The generated torque is compensated :

$$\sum \vec{M_{mr}} = \sum \vec{M_{rr}} \tag{133}$$

find a relation between ω_{mr} and ω_{rr} -> determine the transmission ratio

[2] $\vec{L}$ is perpendicular to both $\vec{P}$ and $\vec{r}$

3.10.3 Pivots à droite et à gauche

pour tourner à gauche ou doite, on ne doit plus satisfaire la condition de stabilité. le pilote utiliser le pédalier pour accélérer/ralentir l'hélice arrière. ainsi les moments cinétiques ne sont plus égaux.

calculer l'effet de rotation sur l'hélicoptère si l'hélice est accélérée/ralentie de 10,20,30,.. $\%$. mettre un tableau. calculer la vitesse de rotation dans ces cas-là.

$$\begin{bmatrix}0\\0\\l_1\end{bmatrix}_{R_1} \quad \vec{AB}_{R_2} = \begin{bmatrix}0\\l_2\\0\end{bmatrix}_{R_2} \quad \vec{BC}_{R_3} = \begin{bmatrix}l_3\\0\\0\end{bmatrix}_{R_3} \quad \vec{CD}_{R_4} = \begin{bmatrix}0\\0\\-l_4\end{bmatrix}_{R_4} \tag{134}$$

.

.

.

$$\begin{aligned}\vec{OE}_R = \vec{OA}_R + \vec{AB}_R + \vec{BB}_{1R} + \vec{B_1C}_{1R} + \vec{C_1C}_R \\ +\vec{CC}_{2R} + \vec{C_2D}_R + \vec{DD}_{1R} + \vec{D_1E}_R\end{aligned} \tag{135}$$

$$\begin{aligned}\vec{OF}_R = \vec{OA}_R + \vec{AB}_R + \vec{BB}_{1R} + \vec{B_1C}_{1R} + \vec{C_1C}_R \\ +\vec{CC}_{2R} + \vec{C_2D}_R + \vec{DD}_{1R} + \vec{D_1E}_R + \vec{EF}_R\end{aligned} \tag{136}$$

$$\begin{aligned}\vec{OG}_R = \vec{OA}_R + \vec{AB}_R + \vec{BB}_{1R} + \vec{B_1C}_{1R} + \vec{C_1C}_R + \vec{CC}_{2R} \\ +\vec{C_2D}_R + \vec{DD}_{1R} + \vec{D_1E}_R + \vec{EF}_R + \vec{FF}_{3R} + \vec{F_3G}_R\end{aligned} \tag{137}$$

$$\begin{aligned}\vec{OH}_R = \vec{OA}_R + \vec{AB}_R + \vec{BB}_{1R} + \vec{B_1C}_{1R} + \vec{C_1C}_R + \vec{CC}_{2R} + \vec{C_2D}_R \\ +\vec{DD}_{1R} + \vec{D_1E}_R + \vec{EF}_R + \vec{FF}_{3R} + \vec{F_3G}_R + \vec{GH}_R\end{aligned} \tag{138}$$

3.11 Conclusion

4 (

Dynamics)

4.1 External forces

If we consider the effect of Earth rotation on a moving object Fictitious tangential force (Euler) : when the a non- uniformly reference frame is used Coriolis Centrifugal force

$$\vec{F'} = \vec{F} - \vec{F}_{Euler} - \vec{F}_{Coriolis} - \vec{F}_{CF} \tag{139}$$

Euler Force

$$\vec{F}_{Euler} = -m\frac{d\vec{\omega}}{dr} * \vec{r} \tag{140}$$

Coriolis Force

$$\vec{F}_{Coriolis} = -2m\vec{\omega} * \vec{v} = -2m\vec{\omega} * \frac{d\vec{r}}{dt} \tag{141}$$

Centrifugal force

$$\vec{F}_{CF} = -m\vec{\omega} * (\vec{\omega} * \vec{r}) \tag{142}$$

* is the operator for the cross product